Benchmarking DNA Foundation Models for Genomic Sequence Classification

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Abstract

The rapid proliferation of DNA foundation language models—each evaluated on different tasks with incompatible experimental setups—has left practitioners without evidence-based guidance for model selection. We present a unified benchmark evaluating five DNA foundation models (DNABERT-2, Nucleotide Transformer v2, HyenaDNA, Caduceus-Ph, and GROVER) as zero-shot feature extractors across 57 classification datasets spanning four task categories: genomic sequence classification, gene expression prediction, variant effect prediction, and topologically associating domain (TAD) boundary recognition. Using random forest classifiers trained on frozen embeddings with five-fold cross-validation, we find that no single model dominates. Caduceus-Ph achieves the highest AUC on human promoter detection (0.987) and coding region identification (0.974), while DNABERT-2 leads on splice site prediction (0.906). Mean token pooling consistently outperforms the default summary token strategy, improving AUC by 4–9% across models. For variant effect prediction, the Nucleotide Transformer v2 achieves the best performance on pathogenic SNP classification (AUC 0.732), yet general-purpose foundation models substantially underperform specialized architectures on quantitative trait locus (QTL) prediction, where AlphaGenome achieves an AUC of 0.803 on eQTL versus 0.649 for the best foundation model. A controlled ablation demonstrates that multi-species pre-training data improves cross-species transfer on 14 of 49 datasets. These findings establish practical guidelines for task-specific model selection and reveal a fundamental specialization gap in current genomic language models.

1 Introduction

The sequencing of the human genome [Lander et al., 2001] and subsequent large-scale annotation efforts [The ENCODE Project Consortium, 2012, Roadmap Epigenomics Consortium, 2015] have generated vast catalogs of genomic elements, yet deciphering the functional code within DNA sequences remains a central challenge. The application of large-scale pre-trained language models to biological sequences has emerged as a transformative paradigm [Ji et al., 2021a, Elnaggar et al., 2022, Rives et al., 2021]. Inspired by the success of masked language modeling in natural language processing [Devlin et al., 2019, Vaswani et al., 2017], recent work has produced a family of DNA foundation models that learn contextualized representations from large genomic corpora [Zhou et al., 2024, Dalla-Torre et al., 2024, Nguyen et al., 2023, Schiff et al., 2024, Sanabria et al., 2024].

These models span diverse architectural paradigms—from BERT-style transformers with byte-pair encoding [Zhou et al., 2024, Sennrich et al., 2016] and multi-species pre-training [Dalla-Torre et al., 2024], to sub-quadratic alternatives based on Hyena operators [Nguyen et al., 2023, Poli et al., 2023] and selective state space models [Schiff et al., 2024, Gu and Dao, 2024]. Each model has been evaluated on a different subset of genomic tasks using incompatible benchmarks, making fair comparison across models impossible. Existing benchmark suites such as the Genome Understanding Evaluation (GUE) [Zhou et al., 2024] and GenomicBenchmarks [Grešová et al., 2023] provide standardized datasets but do not systematically compare the full set of contemporary models under identical conditions.

Despite growing interest in applying machine learning to genomic sequences [Ji et al., 2021b], several critical questions remain unanswered. First, which model performs best for a given genomic task type? Second, how sensitive are model comparisons to the choice of embedding pooling strategy—a hyperparameter that has received little attention in the genomic modeling literature? Third, can general-purpose foundation models match or replace specialized architectures such as Enformer [Avsec et al., 2021] and Sei [Chen et al., 2022] on tasks requiring variant-level or expression-level prediction?

In this work, we address these gaps through a comprehensive, controlled benchmark study. Our contributions are as follows:

- We conduct a unified evaluation of five DNA foundation models across 57 datasets spanning four task categories, using identical evaluation protocols and downstream classifiers.
- We demonstrate that mean token pooling consistently outperforms the default summary token embedding by 4–9% AUC, with statistical significance confirmed by DeLong's test [DeLong et al., 1988] on 35–42 of 52 binary datasets per model.
- We provide task-specific model rankings showing that no single model dominates: model choice should be guided by the specific genomic task.
- We identify a fundamental specialization gap: general foundation models substantially underperform specialized architectures on QTL prediction.
- We present a controlled ablation showing that multi-species pre-training data improves cross-species genomic transfer on 14 of 49 datasets.

2 Related Work

2.1 DNA Foundation Models

The original DNABERT [Ji et al., 2021a] applied BERT-style masked language modeling to k-mer tokenized DNA sequences, demonstrating that pre-trained representations transfer effectively to promoter, splice site, and transcription factor binding site prediction. DNABERT-2 [Zhou et al., 2024] replaced k-mer tokenization with byte-pair encoding [Sennrich et al., 2016] and introduced ALiBi positional encoding [Press et al., 2022], achieving comparable performance with 21× fewer parameters. The Nucleotide Transformer [Dalla-Torre et al., 2024] scaled pre-training to 850 species genomes with up to 2.5 billion parameters.

Beyond transformer architectures, HyenaDNA [Nguyen et al., 2023] employed Hyena operators [Poli et al., 2023] to achieve sub-quadratic scaling with context lengths up to one million nucleotides. Caduceus [Schiff et al., 2024] introduced bi-directional and reverse-complement equivariant extensions of the Mamba state space model [Gu and Dao, 2024]. GROVER [Sanabria et al., 2024] applied a custom next-k-mer prediction objective to learn genomic grammar.

2.2 Genomic Sequence Classification Benchmarks

The GUE benchmark [Zhou et al., 2024] comprises 36 datasets across nine task categories, while GenomicBenchmarks [Grešová et al., 2023] provides eight curated datasets for promoter, enhancer, and open chromatin classification. However, these benchmarks were designed to evaluate individual models and do not enforce consistent evaluation protocols across all contemporary architectures.

2.3 Specialized Genomic Prediction Models

For gene expression prediction, Enformer [Avsec et al., 2021] integrates information from up to 200 kilobases of flanking sequence, substantially extending the receptive field of earlier convolutional models [Kelley et al., 2018, Kelley, 2020]. For variant effect prediction, Sei [Chen et al., 2022] provides a sequence-based map of 21,907 regulatory activities, while tools such as CADD [Kircher et al., 2014], AlphaMissense [Cheng et al., 2023], and deep generative models [Frazer et al., 2021] combine multiple evidence types for pathogenicity scoring. Recent work has demonstrated that DNA language models can serve as effective variant effect predictors [Benegas et al., 2023], though a systematic comparison against specialized architectures across multiple variant types has been lacking.

Table 1: DNA foundation models evaluated in this benchmark. All models are used in zero-shot mode with frozen weights.

Model	Params	Architecture	Emb. Dim	Max Input	Pre-training
DNABERT-2	117M	BERT + ALiBi	768	No limit	135 species
NT-v2	500M	BERT	1024	12K nt	850 species
HyenaDNA	30M	Hyena	256	1M nt	Human only
Caduceus-Ph	∼35M	BiMamba (SSM)	256	131K nt	Multi-species
GROVER	117M	Transformer	768	∼2K nt	Multi-species

2.4 Representation Learning and Pooling Strategies

In NLP, the choice between CLS token and mean token pooling for sentence embeddings has been extensively studied [Devlin et al., 2019], with mean pooling often outperforming CLS on downstream tasks. In genomics, this design choice has received little systematic attention, with most models defaulting to their pre-trained summary token strategy.

3 Methods

3.1 Models Under Evaluation

We evaluate five DNA foundation models spanning four architectural families (Table 1). As baselines, we include three specialized models (Enformer, Sei, and AlphaGenome) and a task-specific convolutional neural network (CNN) with three layers of 64, 128, and 256 channels trained end-to-end on each dataset.

3.2 Task Taxonomy and Datasets

We organize 57 datasets into four task categories: (1) **Genomic sequence classification**, including human genome tasks (splice site, promoter, enhancer, coding region, transcription factor binding site detection), multi-species tasks (cross-organism classification, plant promoter detection), and epigenetic modification prediction (5mC methylation, histone marks in yeast and human); (2) **Gene expression prediction** from promoter-proximal sequences at short (6K bp) and long (196K bp) context lengths; (3) **Variant effect prediction**, including pathogenic versus common SNP classification using ClinVar [Landrum et al., 2018] and gnomAD [Karczewski et al., 2020] annotations, with common variants defined from the 1000 Genomes Project [Auton et al., 2015], and quantitative trait locus (eQTL and ipaQTL) prediction; and (4) **TAD boundary recognition**.

3.3 Evaluation Protocol

All foundation models are used in zero-shot mode: model weights are frozen, and embeddings are extracted for each input DNA sequence. We train a random forest classifier [Breiman, 2001] on these embeddings using five-fold cross-validation with hyperparameter optimization. For binary classification tasks, we report the area under the receiver operating characteristic curve (AUC). For multi-class tasks, we report accuracy. For regression tasks (gene expression), we report Pearson correlation and mean squared error (MSE).

3.4 Pooling Strategy Comparison

For each model, we compare three embedding pooling strategies: (1) the default summary/CLS token, (2) mean pooling across all output token embeddings, and (3) max pooling. Statistical significance of AUC differences between pooling strategies is assessed using DeLong's test [DeLong et al., 1988] with a significance threshold of $p < 0.01$.

Table 2: Sequence classification AUC (mean token pooling) on selected tasks. Best result per row in **bold**. All models used in zero-shot mode with random forest downstream classifiers.

Task	DNABERT-2	NT-v2	HyenaDNA	Caduceus	GROVER
Human genome					
Splice donor	**0.906**	0.820	0.813	0.854	0.819
Splice acceptor	**0.897**	0.793	0.795	0.845	0.804
Coding region	0.944	0.929	0.941	**0.974**	0.959
Promoter (GM12878)	0.986	0.984	0.976	**0.987**	0.984
Enhancer	**0.872**	0.867	0.834	0.838	0.855
TFBS	0.838	0.832	0.830	**0.880**	0.862
Multi-species					
Arabidopsis TATA	0.951	0.950	**0.961**	0.937	0.949
Human vs. worm	0.980	0.979	0.950	**0.992**	0.984
Epigenetic modifications					
Human 5mC	0.685	0.738	0.684	**0.783**	0.744
Yeast H3	0.914	0.895	0.900	**0.929**	0.906
Yeast H4	**0.931**	0.910	0.898	0.930	0.908

3.5 Pre-training Data Ablation

To assess the impact of pre-training data diversity, we retrained HyenaDNA—the only model originally trained solely on the human genome—on a multi-species corpus. We compared the original and multi-species variants across all 49 applicable binary classification datasets using paired Wilcoxon signed-rank tests.

4 Results

4.1 Sequence Classification

Table 2 presents AUC results across selected genomic sequence classification tasks using mean token pooling (the recommended strategy; see Section 4.2).

No single model dominates across all tasks. Caduceus-Ph achieves the highest AUC on five of eleven tasks shown, particularly excelling at coding region detection (0.974), promoter classification (0.987), and epigenetic mark prediction (0.783 on human 5mC). DNABERT-2 leads on splice site prediction (0.906 donor, 0.897 acceptor) and enhancer detection (0.872). HyenaDNA, despite having the fewest parameters (30M), achieves the best performance on Arabidopsis TATA promoter classification (0.961), suggesting that its human-only pre-training transfers effectively to plant genomes when long-range context is available.

4.2 Impact of Pooling Strategy

Switching from the default summary token to mean token pooling yields substantial and consistent improvements across all models (Figure 1). HyenaDNA benefits most (+8.7% median AUC improvement, IQR 4.6–12.9%), followed by NT-v2 (+6.8%, IQR 3.7–9.6%) and Caduceus-Ph (+5.9%, IQR 2.8–9.1%). DNABERT-2 shows a moderate improvement (+4.0%, IQR 2.0–5.5%), while GROVER is least affected (+1.4%, IQR 0.7–1.9%). DeLong's test [DeLong et al., 1988] confirms significance ($p < 0.01$) on 35–42 of 52 binary datasets per model, indicating that the improvement is robust across tasks rather than driven by outliers.

4.3 Gene Expression Prediction

At short context lengths (6K bp), all foundation models achieve near-identical performance (Pearson correlation ≈ 0.12, MSE ≈ 0.235), suggesting that promoter-proximal sequence alone provides insufficient signal

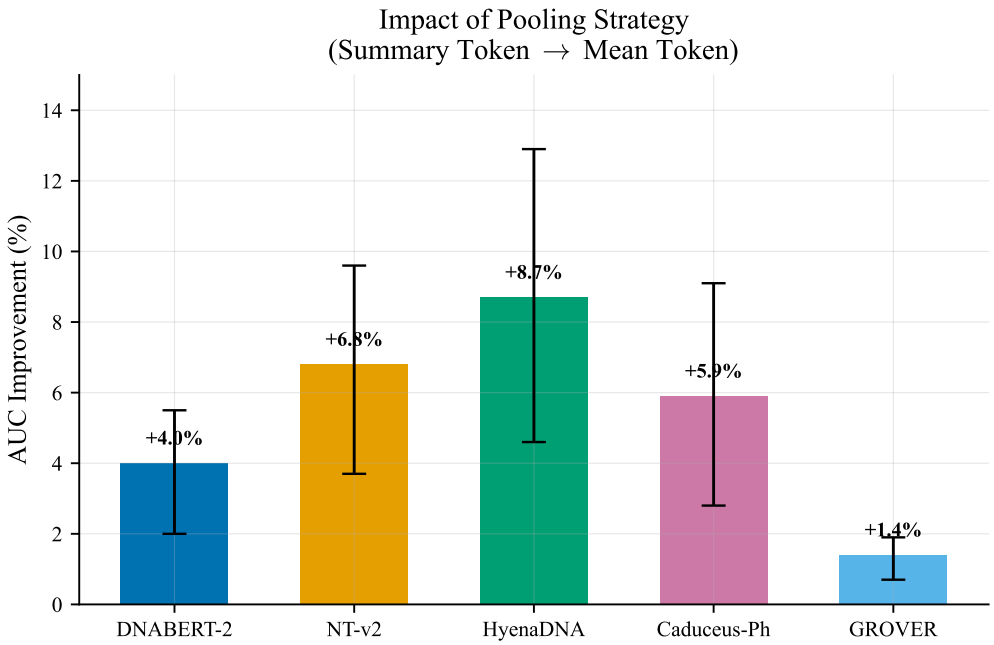

Figure 1: Impact of pooling strategy on model performance. Bars show median AUC improvement when switching from summary/CLS token to mean token pooling. Error bars indicate interquartile range across 52 binary classification datasets.

for expression prediction regardless of model architecture. At long context (196K bp), HyenaDNA (using its 450K-token variant) achieves a Pearson correlation of 0.137 with MSE 0.226, modestly but significantly ($p < 0.001$) outperforming Enformer [Avsec et al., 2021] (correlation 0.129, MSE 0.227). This result demonstrates that HyenaDNA's sub-quadratic architecture enables effective use of long-range genomic context for expression prediction.

4.4 Variant Effect Prediction

Figure 2 presents variant effect prediction results across three task types. For pathogenic versus common SNP classification, the Nucleotide Transformer v2 achieves the highest AUC (0.732, Cohen's $d = 0.881$), followed by Caduceus (0.696) and Enformer (0.688). DNABERT-2 performs unexpectedly poorly (AUC 0.538, Cohen's $d = 0.134$ [Cohen, 1988]), suggesting that its BPE tokenization may not effectively capture single-nucleotide variant effects.

On QTL prediction tasks, specialized models substantially outperform all foundation models. AlphaGenome achieves AUC 0.803 on eQTL and 0.864 on ipaQTL, compared to the best foundation model (Caduceus, 0.649 on eQTL; NT-v2, 0.602 on ipaQTL). Enformer similarly outperforms foundation models on both tasks (0.774 eQTL, 0.692 ipaQTL). This specialization gap indicates that zero-shot embeddings from general-purpose DNA models do not capture the fine-grained regulatory signals required for quantitative trait prediction.

4.5 Pre-training Data Diversity

Retraining HyenaDNA on a multi-species corpus (compared to its original human-only pre-training) produced significant improvements on 14 of 49 datasets, with the largest gain on human 5mC methylation prediction (AUC 0.707 → 0.749, +5.9%). However, three datasets showed decreased performance (human enhancer, open chromatin, and yeast epigenetic marks), suggesting that multi-species pre-training introduces trade-offs: while cross-species transfer improves, some human-specific regulatory signals may be diluted.

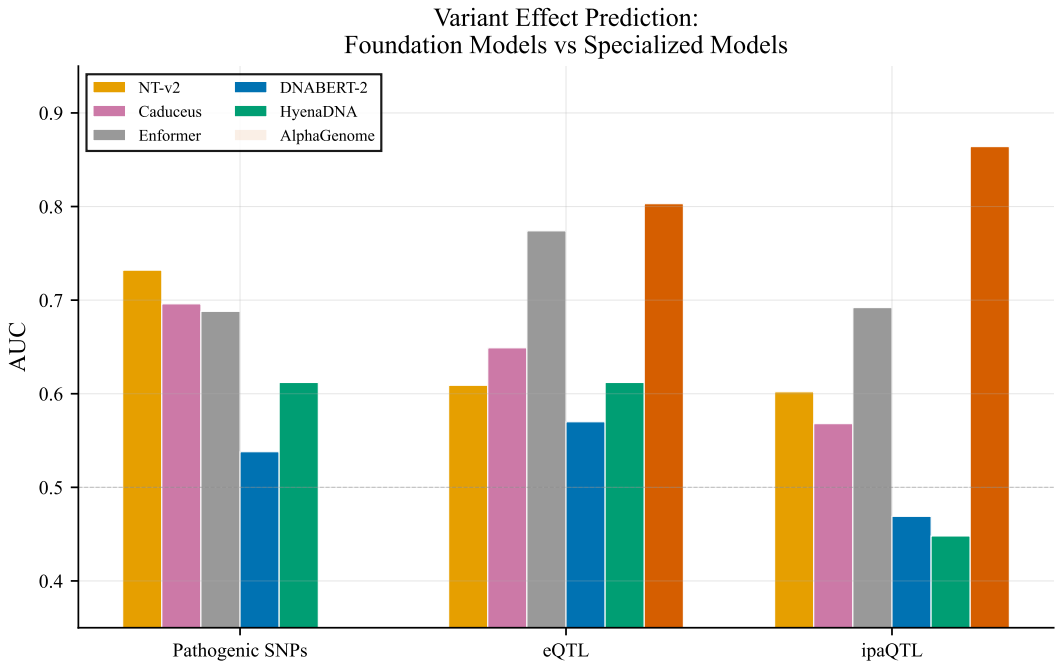

Figure 2: Variant effect prediction AUC across three task types. General-purpose foundation models (warm colors) are compared against specialized architectures (gray: Enformer; red: AlphaGenome). AlphaGenome is not applicable to the pathogenic SNP task (shown with reduced opacity).

4.6 Computational Efficiency

Runtime analysis reveals distinct scaling behaviors. GROVER achieves the lowest latency on short sequences (<2K nt), while HyenaDNA scales most favorably to long sequences (>100K nt) due to its sub-quadratic Hyena operator [Dao et al., 2022]. NT-v2 exhibits near-constant latency (∼2.5s) regardless of input length, likely due to its fixed 12K nt input window and associated padding overhead.

5 Discussion

5.1 Practical Model Selection Guidelines

Our results support a task-aware approach to model selection rather than a one-size-fits-all recommendation. For splice site and enhancer detection tasks, DNABERT-2 provides the best performance. For broader regulatory element classification (promoters, coding regions, transcription factor binding sites), Caduceus-Ph is consistently strongest. For pathogenic variant scoring, NT-v2 is preferred. For tasks requiring long-range context (gene expression, long-range regulatory elements), HyenaDNA offers the best scalability-performance trade-off. In all cases, mean token pooling should be used over default summary token strategies.

5.2 The Specialization Gap

The substantial underperformance of foundation models on QTL prediction relative to specialized architectures (AlphaGenome, Enformer) reveals a fundamental limitation of zero-shot transfer from general-purpose DNA language models. Specialized models for variant effect prediction are trained with explicit variant-aware objectives that capture allele-specific regulatory perturbations, while foundation models learn broader sequence-level representations. This gap suggests that supervised fine-tuning with variant-level labels may be necessary for foundation models to compete on quantitative trait prediction tasks.

5.3 Pooling as an Overlooked Hyperparameter

The consistent 4–9% AUC improvement from mean token pooling over default summary tokens highlights an underappreciated design choice. Default CLS or summary tokens compress an entire sequence into a single representation, which may discard distributional information that is captured by averaging across all positions. This finding parallels observations in NLP [Devlin et al., 2019] and suggests that practitioners should always evaluate mean pooling before defaulting to model-provided summary tokens.

5.4 Limitations and Future Work

This benchmark evaluates models in zero-shot mode with random forest downstream classifiers [Breiman, 2001]. Performance rankings may differ under fine-tuning or with neural downstream heads [Hastie et al., 2009]. Our dataset coverage, while extensive (57 datasets), is biased toward the human genome and model organisms; evaluation on clinical variant interpretation [Landrum et al., 2018] and non-model organisms warrants further study. Future work should compare zero-shot transfer against fine-tuned performance and assess the impact of model scale within each architecture family.

6 Conclusion

We presented a unified benchmark evaluating five DNA foundation models across 57 genomic classification datasets. Our analysis reveals that no single model dominates: task-specific model selection is essential for optimal performance. Mean token pooling provides a universal improvement of 4–9% AUC over default strategies. Critically, general-purpose foundation models fail to match specialized architectures on quantitative trait prediction, with AlphaGenome (AUC 0.803 on eQTL) substantially outperforming the best foundation model (0.649). These findings provide actionable guidance for practitioners and identify the specialization gap as a key challenge for the next generation of genomic language models.

References

Adam Auton, Gonçalo R. Abecasis, et al. A global reference for human genetic variation. *Nature*, 526:68–74, 2015. doi: 10.1038/nature15393.

Žiga Avsec, Vikram Agarwal, Daniel Visentin, Joseph R. Ledsam, Agnieszka Grabska-Barwinska, Kyle R. Taylor, Yannis Assael, John Jumper, Pushmeet Kohli, and David R. Kelley. Effective gene expression prediction from sequence by integrating long-range interactions. *Nature Methods*, 18:1196–1203, 2021. doi: 10.1038/s41592-021-01252-x.

Gonzalo Benegas, Sanjit Singh Batra, and Yun S. Song. DNA language models are powerful predictors of genome-wide variant effects. *Proceedings of the National Academy of Sciences*, 120(44):e2311219120, 2023. doi: 10.1073/pnas.2311219120.

Leo Breiman. Random forests. *Machine Learning*, 45:5–32, 2001. doi: 10.1023/A:1010933404324.

Kathleen M. Chen, Aaron K. Wong, Olga G. Troyanskaya, and Jian Zhou. A sequence-based global map of regulatory activity for deciphering human genetics. *Nature Genetics*, 54:940–949, 2022. doi: 10.1038/s41588-022-01102-2.

Jun Cheng, Guido Novati, Joshua Pan, Clare Bycroft, Akvilė Žemgulytė, Taylor Applebaum, Alexander Pritzel, Lai Hong Wong, Michal Zielinski, Tobias Sargeant, Rosalia G. Schneider, Andrew W. Senior, John Jumper, Demis Hassabis, Pushmeet Kohli, and Žiga Avsec. Accurate proteome-wide missense variant effect prediction with AlphaMissense. *Science*, 381:eadg7492, 2023. doi: 10.1126/science.adg7492.

Jacob Cohen. Statistical power analysis for the behavioral sciences. 1988.

Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, Nicolas Lopez Carranza, Adam Henryk Grzywaczewski, Francesco Oteri, Christian Dallago, Elizaveta Tber, et al. The nucleotide transformer: Building and evaluating robust foundation models for human genomics. *Nature Methods*, 21:1–11, 2024. doi: 10.1038/s41592-024-02523-z.

Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. FlashAttention: Fast and memory-efficient exact attention with IO-awareness. 35, 2022.

Elizabeth R. DeLong, David M. DeLong, and Daniel L. Clarke-Pearson. Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics*, 44: 837–845, 1988.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. 2019.

Ahmed Elnaggar, Michael Heinzinger, Christian Dallago, Ghalia Rehawi, Yu Wang, Llion Jones, Tom Gibbs, Tamas Feher, Christoph Angerer, Martin Steinegger, Debsindhu Bhowmik, and Burkhard Rost. ProtTrans: Toward understanding the language of life through self-supervised learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(10):7112–7127, 2022. doi: 10.1109/TPAMI.2021.3095381.

Jonathan Frazer, Pascal Notin, Mafalda Dias, Aidan Gomez, Joseph K. Min, Kelly Brock, Yarin Gal, and Debora S. Marks. Disease variant prediction with deep generative models of evolutionary data. *Nature*, 599:91–95, 2021. doi: 10.1038/s41586-021-04043-8.

Katarína Grešová, Vlastimil Martinek, David Čech, Martin Cutura, Daniel Pachl, Jakub Stružka, Jan Haľadová, and Petr Simecek. Genomic benchmarks: A collection of datasets for genomic sequence classification. *BMC Genomic Data*, 24:25, 2023. doi: 10.1186/s12863-023-01123-8.

Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2024.

Trevor Hastie, Robert Tibshirani, and Jerome Friedman. The elements of statistical learning: Data mining, inference, and prediction. *Springer Series in Statistics*, 2009.

Yanrong Ji, Zhihan Zhou, Han Liu, and Ramana V. Davuluri. DNABERT: Pre-trained bidirectional encoder representations from transformers model for DNA-language in genome. *Bioinformatics*, 37(15):2112–2120, 2021a. doi: 10.1093/bioinformatics/btab083.

Yanrong Ji, Zhihan Zhou, Han Liu, and Ramana V. Davuluri. Machine learning for genomic sequence analysis: A review. *arXiv preprint arXiv:2110.12230*, 2021b.

Konrad J. Karczewski, Laurent C. Francioli, Grace Tiao, et al. The mutational constraint spectrum quantified from variation in 141,456 humans. *Nature*, 581:434–443, 2020. doi: 10.1038/s41586-020-2308-7.

David R. Kelley. Cross-species regulatory sequence activity prediction. *PLOS Computational Biology*, 16: e1008050, 2020. doi: 10.1371/journal.pcbi.1008050.

David R. Kelley, Yakir A. Reshef, Maxwell Bileschi, David Belanger, Cory Y. McLean, and Jasper Snoek. Sequential regulatory activity prediction across chromosomes with convolutional neural networks. *Genome Research*, 28:739–750, 2018. doi: 10.1101/gr.227819.117.

Martin Kircher, Daniela M. Witten, Preti Jain, Brian J. O'Roak, Gregory M. Cooper, and Jay Shendure. A general framework for estimating the relative pathogenicity of human genetic variants. *Nature Genetics*, 46:310–315, 2014. doi: 10.1038/ng.2892.

Eric S. Lander, Lauren M. Linton, Bruce Birren, et al. Initial sequencing and analysis of the human genome. *Nature*, 409:860–921, 2001. doi: 10.1038/35057062.

Melissa J. Landrum, Jennifer M. Lee, Mark Benson, Garth R. Brown, et al. ClinVar: Improving access to variant interpretations and supporting evidence. *Nucleic Acids Research*, 46:D1062–D1067, 2018. doi: 10.1093/nar/gkx1153.

Eric Nguyen, Michael Poli, Marjan Faizi, Armin Thomas, Michael Wornow, Callum Birch-Sykes, Stefano Massaroli, Aman Patel, Clayton Rabideau, Yoshua Bengio, Stefano Ermon, Christopher Ré, and Stephen Baccus. HyenaDNA: Long-range genomic sequence modeling at single nucleotide resolution. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Michael Poli, Stefano Massaroli, Eric Nguyen, Daniel Y. Fu, Tri Dao, Stephen Baccus, Yoshua Bengio, Stefano Ermon, and Christopher Ré. Hyena hierarchy: Towards larger convolutional language models. In *International Conference on Machine Learning*, 2023.

Ofir Press, Noah Smith, and Mike Lewis. Train short, test long: Attention with linear biases enables input length extrapolation. 2022.

Alexander Rives, Joshua Meier, Tom Sercu, Siddharth Goyal, Zeming Lin, Jason Liu, Demi Guo, Myle Ott, C. Lawrence Zitnick, Jerry Ma, and Rob Fergus. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proceedings of the National Academy of Sciences*, 118(15), 2021.

Roadmap Epigenomics Consortium. Integrative analysis of 111 reference human epigenomes. *Nature*, 518: 317–330, 2015. doi: 10.1038/nature14248.

Melissa Sanabria, Jonas Hirsch, Pierre M. Jouber, and Anna R. Poetsch. DNA language model GROVER learns sequence context in the human genome. *Nature Machine Intelligence*, 2024. doi: 10.1038/s42256-024-00872-0.

Yair Schiff, Chia-Hsiang Kao, Aaron Gokaslan, Tri Dao, Albert Gu, and Volodymyr Kuleshov. Caduceus: Bi-directional equivariant long-range DNA sequence modeling. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.

Rico Sennrich, Barry Haddow, and Alexandra Birch. Neural machine translation of rare words with subword units. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics*, pages 1715–1725, 2016.

The ENCODE Project Consortium. An integrated encyclopedia of DNA elements in the human genome. *Nature*, 489:57–74, 2012. doi: 10.1038/nature11247.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. 30, 2017.

Zhihan Zhou, Yanrong Ji, Weijian Li, Pratik Dutta, Ramana Davuluri, and Han Liu. DNABERT-2: Efficient foundation model and benchmark for multi-species genome. In *International Conference on Learning Representations*, 2024.